

Ubiquitous Well-being:

Patchable AI Unites Computing and Psychological Science

Xingyu Li

Department of Psychology
Institute for Human-centered AI
Stanford University
Stanford, California, USA
axyli@stanford.edu

Xianglu Tang

Institute for Human-centered AI
Stanford University
Stanford, California, USA
xianglutang111@gmail.com

Emily Ma

Institute for Human-centered AI
Stanford University
Stanford, California, USA
emilylihanma88@gmail.com

Geoffrey Cohen

Graduate School of Education
Department of Psychology
Stanford University
Stanford, California, USA
glc@stanford.edu

Abstract

Brief psychological interventions (e.g., values affirmation) improve health and learning, yet an infrastructure for personalized, scalable, and timely delivery is missing. We propose that ubiquitous computing can transform static interventions into evolving services. This work introduces Patchable AI, a framework in which chatbots become continuously learning systems through hot-swappable, auditable “intervention patches” guarded by safety policies. Four architectural layers – sensor triggers, policy guardrails, patch registry, and conversation broker – allow evolution without retraining underlying models. To provide materials for brief psychological interventions, we conducted a large-scale data collection effort using a custom web-scriber to capture nearly all publicly available psychological and journaling prompts ($N = 54,251$) from 2015 onward, yielding short, crisp contents structured for patchable deployment. Our values-affirmation case study demonstrates how Patchable AI aligns psychological theory, organizational policy, and technical affordances, offering a roadmap for scalable, personalized well-being services. This research aims to (1) apply a ubiquitous-computing framework across disciplines—education, public health, workplace training—and (2) advance intervention science by embedding human-centered AI that turns static protocols into adaptive, context-aware services. We also outline broader implications, showing how Patchable AI can underpin living support systems across domains.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing → Context-aware computing • Software and its engineering → Software architectures • Applied computing → Health informatics

Keywords

Patchable AI, Psychological Interventions, Chatbots

ACM Reference format:

Xingyu Li, Xianglu Tang, Emily Ma, and Geoffrey Cohen. 2025. Ubiquitous Well-being: Patchable AI Unites Computing and Psychological Science. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3754377>

1 Introduction

For decades, research shows that brief, timely psychological interventions—such as writing a gratitude journal, reflecting on personal values, practicing a mindful deep breath, or reframing a setback as something temporary and normal—can measurably improve health, learning, and well-being [1]. In addition, a large body of evidence converges on a key design principle: these interventions work best when they are timely, targeted, and tailored—delivered at the right moment to the right person in the right tone [2, 3]. These interventions hold great promise, and large-scale implementations do exist—for example, Yeager’s work on growth mindset [4] or Borman’s studies on affirmation [5]. However, there is still no widely deployed approach that delivers interventions that are both scalable and personalized. Advances in AI and ubiquitous computing now make this vision newly achievable.

AI adoption is accelerating at unprecedented speed. According to the Stanford Institute for Human-Centered AI’s 2025 AI Index, global investment in AI exceeded \$250 billion last year, and the number of AI-related job postings has more than doubled since 2022. Chatbots, in particular, have emerged as the dominant interface: Gartner projects that by 2026, 95% of customer interactions will be mediated by AI-powered agents, while enterprise software platforms such as Salesforce and Microsoft now integrate conversational assistants as default features [6,7].

This shift is not limited to the private sector. Educational institutions, healthcare systems, and public agencies increasingly rely on AI assistants for student advising, patient screening, and benefits navigation—bringing Weiser’s vision of ubiquitous computing into everyday life [8,9]. Modern institutional chatbots—already embedded as “digital employees” in schools, hospitals, and workplaces—can fuse ambient cues such as time, location, calendar

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp Companion '25, October 12–16, 2025, Espoo, Finland

© 2025 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1477-1/25/10

<https://doi.org/10.1145/3714394.3754377>

events, keystroke rhythms, and wearable biomarkers to infer a student’s midnight exam anxiety, a patient’s pre-operative stress, or an employee’s rising fatigue, then respond with precisely personalized support seconds later [10, 11, 12]. This ubiquity unlocks, for the first time, the prospect of delivering genuinely targeted, timely, and psychologically meaningful interventions at population scale.

2 Related Work

Prior work has laid important groundwork by developing systems that deliver support at key moments. For example, ubicomp researchers have pioneered just-in-time adaptive interventions (JITAI) [14], mobile sensing, and mHealth tools that trigger support when GPS, accelerometers or self-reports indicate a psychologically meaningful moment [12, 15]. While these systems effectively sense when help is needed, their behavioral scripts are typically hard-coded, leading to “deployment decay” as policies or risk factors change.

Psychologists, psychiatrists, and behavioral scientists have also long sought to deliver support at the right moment—using a wide range of methods like email check-ins, self-help books, chatbots, one-on-one therapy, mobile mental health apps, and online support groups. As outlined in Figure 1, each of these approaches brings strengths, but also serious tradeoffs. For instance, one-on-one therapy offers highly personalized, expert guidance, but it is resource-intensive and difficult to scale to large populations.

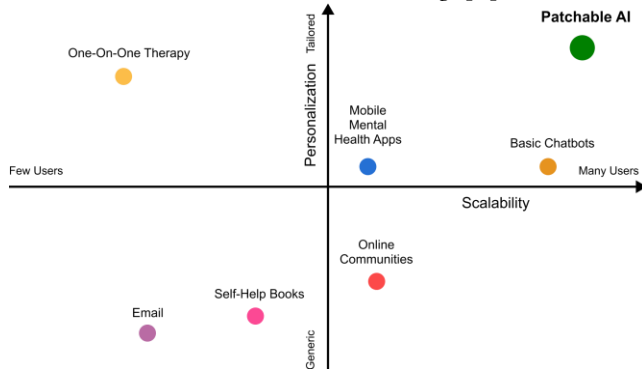


Figure 1: Design Space for Scalable Intervention Delivery

On the other end of the spectrum, methods like email campaigns or mobile health apps can reach many users at once, but they often rely on static content and lack the nuance to adapt to a person’s specific situation in real time. Even the latest large language models—despite their ability to generate fluent, engaging dialogue—are not optimized for delivering psychological interventions. They lack deep domain expertise in behavioral science, have no guaranteed access to a user’s full context, and cannot iteratively adapt support toward long-term mental health outcomes. As a result, they often fall short in delivering structured, science-based interventions that meaningfully evolve over time.

To bridge the gap between scalability and personalization, we propose a new vision: Patchable AI—a modular, auditable framework that enables psychological interventions to be swapped, tuned, or retired as easily as adding a plugin.

3 Patchable AI Framework

Patchable AI turns each psychological intervention into a hot-swappable “patch”—a modular unit that can be injected, rolled back, or re-tuned without modifying the underlying LLM or retraining the core model. The goal is to give psychologists and system administrators fine-grained control over intervention content and timing, while retaining the scalability and flexibility of a chatbot interface. Patchable AI leverages well-established ubicomp principles such as runtime adaptation, context-aware computing, and resilient design to deliver just-in-time support [3, 4, 5, 6]. But unlike typical JITAI or mobile sensing pipelines that focus on “when” support should be given, Patchable AI also emphasizes “what” and “how”—by dynamically selecting from a library of validated psychological patches based on real-time signals, historical patterns, and outcome predictions.

In this section, we present the full framework, outlining its core design goals, system architecture, and how it integrates predictive modeling and user feedback to self-update over time.

3.1 System Overview

Patchable AI consists of five core modules (Fig. 2): (1) Chat Interface, (2) Patch Engine, (3) Contextual Sensing, (4) Predictive Routing, and (5) Feedback Loop. These modules operate in real-time, orchestrated through a lightweight policy controller, and communicate with a centralized Patch Registry that manages available psychological interventions.

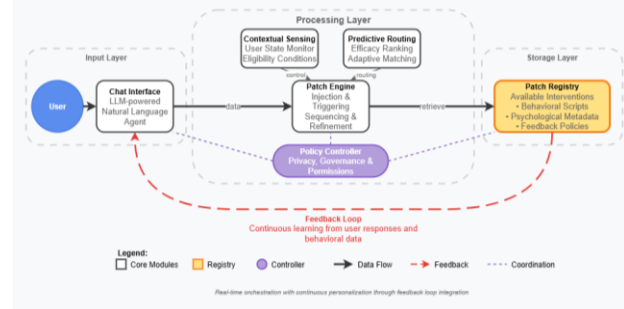


Figure 2: Patchable AI System Architecture

4 Values-Affirmation Patch: A Case Study

To illustrate how Patchable AI works in practice, we walk through a complete deployment pipeline for a single, well-validated intervention: values affirmation. This example demonstrates how a modular “patch” can be inserted into an AI assistant to deliver timely, tailored, and theoretically grounded support in real-world settings.

4.1 Why Values Affirmation?

Values affirmation is a brief psychological intervention in which individuals reflect on their core personal values—such as family, creativity, or perseverance—and articulate why those values are meaningful to them [16]. In self-affirmation theory, the practice helps buffer against psychological threats by reinforcing a stable sense of self-worth and identity [17, 18].

Over the past two decades, values-affirmation interventions have been shown to produce significant, long-lasting benefits across a wide range of domains. In education, short reflection exercises have reduced racial achievement gaps in middle school classrooms [19] and boosted college retention for first-generation students [20]. In health psychology, values-affirmation has improved medication adherence, reduced cortisol responses under stress, and promoted adaptive coping in patients managing chronic illness [21, 22]. Effects are often durable, with some studies reporting benefits that persist months or even years after a single 15-minute writing prompt [22].

4.2 Patch Components

4.2.1 Prompt Corpus and Ontology. To ground the Patchable AI system in real-world, human-authored writing, we constructed a large-scale corpus of values-affirmation and mindfulness-related prompts spanning nearly a decade. This corpus serves as the raw material from which the system draws and classifies interventions. We captured the full public record of English-language mindful journaling prompts published online between January 1, 2016, and June 15, 2025. To do so, we built a four-stage, multi-threaded web crawling pipeline designed for scale, precision, and temporal traceability. Figure 3 illustrates the overall workflow.

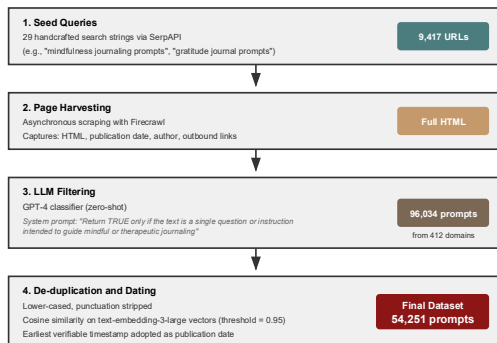


Figure 3: Multi-threaded Web-crawling Pipeline for Mindful-writing Prompt Collection

The final dataset comprises 54,251 unique prompts, each with metadata including domain, timestamp, and source URL. This corpus reflects a diverse cross-section of public psychological interventions, spanning wellness blogs, educational platforms, therapeutic workbooks, and mental health startups.

To organize the dataset for structure-aware retrieval, we labeled each prompt along two intuitive dimensions. The first is temporal focus—whether the prompt directs attention to the past, present, or future. The second is prompt function—the kind of cognitive or emotional action it elicits.

For example, a recall prompt might ask someone to describe a meaningful achievement; a reframe prompt might invite them to view a setback as temporary; an envision prompt might encourage imagining a hopeful future; a clarify prompt could ask

them to articulate their core values; and a connect prompt might guide them to express appreciation for important relationships.

This framework allows the system to select prompts not merely by topic, but by the mental shift most useful in a given moment. Figure 4 illustrates this conceptual map.

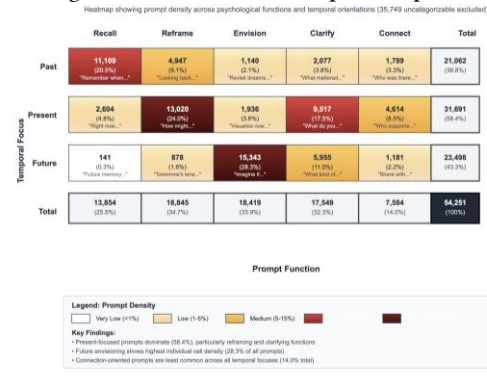


Figure 4: Distribution of 54,251 Categorizable Web-Scraped Journaling Prompts by Function and Temporal Focus.

4.2.2 Values Affirmation (VA) Module Implementation and Application

4.2.2.1 Adapting Values Affirmation with UbiComp Principles. The VA module exemplified the framework's capabilities by transforming a traditional, static psychological intervention into a dynamic, context-aware service. Traditional lab-based VA exercises involve writing about personal values before a stressor (e.g., academic tests). Our implementation leveraged the Patchable AI architecture to embed this principle into an adaptive service.

4.2.2.2 Contextual Triggering of the VA Module. The VA module specifically utilized the Context Awareness Layer by responding to a range of real-world triggers pertinent to self-affirmation needs:

- 1) Temporal triggers: Pre-exam periods, late-night study sessions
- 2) Behavioral triggers: Expressions of academic anxiety or self-doubt
- 3) Environmental triggers: High-stress institutional contexts (exam weeks, evaluation periods)
- 4) Longitudinal triggers: Patterns of recurring stress or diminished self-efficacy

These triggers align with principles of context-awareness in smart environments [7, 8, 16].

4.2.2.3 Personalized Content Delivery in the VA Module. Rather than generic worksheets, the framework delivers personalized affirmations built from users' own disclosed values and experiences. Narrative building algorithms help users construct coherent autobiographical stories that emphasize personal strengths and meaningful accomplishments, while normalization features provide statistical context that validates their experiences [11, 20].

4.2.2.4 Organizational Deployment. The modular design accommodates institutional constraints through configurable

policy guardrails. Schools can specify age-appropriate content filters, cultural sensitivity requirements, and privacy protection protocols without requiring technical expertise. Hot-swappable deployment enables administrators to install, update, or remove intervention modules as institutional needs evolve.

4.3 Conceptual Contributions

4.3.1 UbiComp-Intervention Science Synthesis. Our framework demonstrates that ubicomp's foundational principles directly address core challenges in intervention science:

- Context awareness enables just-in-time delivery of psychological exercises when they're most needed and most likely to be effective [3, 4, 5, 6, 14, 16].
- Run-time adaptation allows personalization at scale through modular intervention patches that evolve with individual and institutional needs [3, 4, 6, 9, 20].
- In-the-wild longevity ensures sustained effectiveness through continuous learning and validation mechanisms.

4.3.2 Scalable Personalization Solution. The design resolves the false dichotomy between scale and contextual fit. By treating chatbots as learners in continuing education, the framework achieves population-level reach while maintaining moment-to-moment responsiveness to individual circumstances and institutional cultures [10, 11, 20].

4.3.3 Advancing Intervention Delivery Infrastructure. Patchable AI creates sustainable infrastructure for psychological intervention delivery. Rather than requiring new technological implementations for each evidence-based exercise, researchers can package interventions as modular patches that integrate into existing institutional chatbot deployments with full audit trails and safety mechanisms. The values affirmation case study proves the framework's practical viability – demonstrating how robust psychological theory, organizational requirements, and technical constraints can be successfully unified within a single, coherent system architecture that honors both scientific rigor and real-world complexity.

Overall, this framework's contributions are significant when contrasted with prior work, which offers only partial solutions. UbiComp researchers have pioneered JITAI [14], mobile sensing platforms, and mHealth toolkits that trigger support when GPS, accelerometers or self-reports indicate a teachable moment [12, 15]. While these systems admirably sense when help is needed, their underlying behavioral scripts are usually hard-coded, leading to "deployment decay" as policies, slang, or risk factors shift and the logic grows stale. Meanwhile, behavioral scientists experimenting with chatbots typically embed single micro-interventions—values-affirmation, expressive-writing, growth-mindset prompts—inside static conversational flows [4, 9]. This results in a patchwork of siloed prototypes: context-aware but frozen, or evidence-based but blind to context, and almost never maintained beyond the initial study period.

Patchable AI introduces the missing modular, auditable update mechanism, allowing interventions to be swapped, tuned, or retired as effortlessly as installing an app. Chatbots, already integrated into schools and hospitals, provide the delivery channel

[16, 19]; ubicomp supplies the design principles for run-time adaptation, guard-railed safety, and in-the-wild resilience [3, 4, 6]. By marrying these traditions, Patchable AI treats each psychological exercise as a hot-pluggable "patch." Administrators can, for instance, add a values-affirmation module before exams, roll it back if it clashes with local norms, or push a mindfulness check-in when semester stress peaks—all without retraining the core language model or disrupting ongoing conversations. In doing so, we offer the first unified approach that simultaneously satisfies the psychologist's mandate for personal, context-tailored delivery and the engineer's demand for scalable, maintainable infrastructure.

5 Discussion

Patchability as a Middle Path. By decoupling user-facing outputs from the base model, Patchable AI offers a third design path—between fine-tuning and prompting—that affords flexibility, oversight, and rapid iteration.

Domain Transferability. Although we focused on values affirmation, the architecture generalizes to other therapeutic, educational, or productivity interventions where personalization and safety are paramount.

Scalability vs. Personalization. A core tension lies in delivering personalized psychological support at population scale without eroding the intimacy and contextual nuance such support demands.

Data Challenges. We rely heavily on prompt-level data, but future iterations must account for multimodal inputs and richer, longitudinal user-state tracking.

Ethical Oversight. Real-time personalization amplifies both benefit and risk. Transparent patch history, administrator controls, and user opt-outs are not optional—they are core features for responsible deployment.

6 Future Directions

Future work will advance the architecture along several fronts. First, we will extend patch processing to multimodal cues—voice tone, facial expressions, and biosignals—so the system can ground its responses in richer contextual signals. Second, the current bandit-based routing will evolve toward full reinforcement-learning frameworks that optimize long-term outcomes such as well-being, habit formation, and sustained goal attainment. Third, cross-session memory structures will be introduced to knit individual prompts into a coherent personal narrative, enabling the agent to track and refine a user's evolving self-concept and objectives. Finally, to test the framework's domain-agnostic promise, we plan pilots in healthcare coaching, teacher dashboards, and workplace well-being, assessing how patchability scales across distinct high-stakes contexts.

References

- [1] Geoffrey L. Cohen, Julio Garcia, Nancy Apfel, and Allison Master. 2006. Reducing the racial achievement gap: A social-psychological intervention. *Science* 313, 5791 (September 2006), 1307–1310. <https://doi.org/10.1126/science.1128317>
- [2] Geoffrey L. Cohen, Julio Garcia, and Jonathan P. Goyer. 2017. Turning point: Targeted, tailored, and timely psychological intervention. In *Handbook of*

- Competence and Motivation* (2nd ed.), Andrew J. Elliot, Carol S. Dweck, and David S. Yeager, Eds., Guilford Press, 657–686.
- [3] Kathleen Kara Fitzpatrick, Alison Darcy, and Molly Vierhile. 2017. Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial. *JMIR Mental Health* 4, 2 (June 2017), e19. <https://doi.org/10.2196/mental.7785>
 - [4] David S. Yeager, Paul Hanselman, Gregory M. Walton, Joshua S. Murray, Robert Crosnoe, Charissa Müller, Erin Tipton, Barrett Schneider, Caitlin S. Hulleman, Carlos P. Hinojosa, David Paunesku, Cara Romero, Katherine Flint, Alexandra Roberts, Joshua Trott, Rebecca Iachan, Jordan Buontempo, Stephanie M. Yang, Cláudia M. Carvalho, Peter R. Hahn, Mathew Gopalan, Padma Mhatre, Russell Ferguson, Angela L. Duckworth, and Carol S. Dweck. 2019. A national experiment reveals where a growth mindset improves achievement. *Nature* 573, 7774 (2019), 364 – 369. <https://doi.org/10.1038/s41586-019-1466-y>
 - [5] Geoffrey D. Borman, Jason Grigg, and Paul Hanselman. 2016. An effort to close achievement gaps at scale through self-affirmation. *Educational Evaluation and Policy Analysis* 38, 1 (2016), 21 – 42. <https://doi.org/10.3102/0162373715581709>
 - [6] McKinsey & Company. 2025. *The State of AI*. McKinsey & Company. Retrieved August 4, 2025 from <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>
 - [7] CloudTalk. 2025. *Customer Interaction Management: AI to Optimize Ops & CX*. CloudTalk Blog. Retrieved August 4, 2025 from <https://www.cloudtalk.io/blog/customer-interaction-management/>
 - [8] Weiser, M. 1991. *The Computer for the 21st Century*. Scientific American 265 (3), 94–104.
 - [9] K. Collins. 2024. *iPhone Designer Jony Ive and OpenAI's Sam Altman Working on Top-Secret AI Device*. CNET. Retrieved August 4, 2025 from <https://www.cnet.com/tech/mobile/iphone-designer-jony-ive-and-openais-sam-altman-working-on-top-secret-ai-device-report-says/>
 - [10] Schillings, C., Meissner, D., Erb, B., Schultchen, D., Bendig, E., & Pollatos, O. 2024. Effects of a Chatbot-Based Intervention on Stress and Health: Randomized Controlled Trial in the Workforce. *JMIR Mental Health* 11 (1), e50454. <https://mental.jmir.org/2024/1/e50454>
 - [11] Ulrich, S., Mettler, T., Jørgensen, A., & Kowatsch, T. 2024. MISHA: A Conversational-Agent Stress-Management Coach for Students. *JMIR mHealth* 12 (1), e54945. <https://mhealth.jmir.org/2024/1/e54945>
 - [12] MedPath Newsroom. 2025. *First Clinical Trial Shows AI Therapy Bot Effective for Depression and Anxiety*. MedPath News. Retrieved August 4, 2025 from <https://trial.medpath.com/news/51d1f8cd380159f8/first-clinical-trial-shows-ai-therapy-bot-effective-for-depression-and-anxiety>
 - [13] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. ACM, New York, NY, USA, 610–623. <https://doi.org/10.1145/3442188.3445922>
 - [14] Inbal Nahum-Shani, Shawna N. Smith, Bonnie J. Spring, Linda M. Collins, Katie Witkiewitz, Ambuj Tewari, and Susan A. Murphy. 2018. Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support. *Annals of Behavioral Medicine* 52, 6 (2018), 446–462. <https://doi.org/10.1007/s12160-016-9830-8>
 - [15] Javier Cámara, Carlos Canal, and Nikolay Vasilev. 2010. A framework for run-time behavioural service adaptation in ubiquitous computing. In *On the Move to Meaningful Internet Systems: OTM 2010 Workshops (OTM '10)*, LNCS 6428, Springer, Berlin, Heidelberg, 67–76. https://doi.org/10.1007/978-3-642-16961-0_15
 - [16] Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. 2024. WildChat: 1M ChatGPT Interaction Logs in the Wild. In *Proceedings of the International Conference on Learning Representations (ICLR 2024)*. Vienna, Austria. arXiv:2405.01470. <https://doi.org/10.48550/arXiv.2405.01470>
 - [17] Claude M. Steele. 1988. The psychology of self-affirmation: Sustaining the integrity of the self. *Advances in Experimental Social Psychology* 21 (1988), 261–302.
 - [18] Geoffrey L. Cohen and David K. Sherman. 2014. Self-affirmation and social psychological intervention. *Annual Review of Psychology* 65 (2014), 333–371. <https://doi.org/10.1146/annurev-psych-010213-115137>
 - [19] Zhiqiu Jiang, Mashrur Rashik, Kunjal Panchal, Mahmood Jasim, Ali Sarvghad, Pari Riahi, Erica Dewitt, Fey Thurber, and Narges Mahyar. 2023. CommunityBots: Creating and Evaluating A Multi-Agent Chatbot Platform for Public Input Elicitation. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1, Article 36 (April 2023), 32 pages. <https://doi.org/10.1145/3579469>
 - [20] Bahar Irfan, Michael Garcia Ortiz, Natalia Lyubova, and Tony Belpaeme. 2021. Multi-modal open world user identification. *ACM Transactions on Human-Robot Interaction* 11, 1 (October 2021), Article 6. <https://doi.org/10.1145/3477963>
 - [21] Samuele Centorrino, Jean-Pierre Florens, and Jean-Michel Loubes. 2022. Fairness constraint in structural econometrics and application to fair estimation using instrumental variables. *arXiv preprint arXiv:2202.08977*. <https://doi.org/10.48550/arXiv.2202.08977>
 - [22] Frank Kaptein, Bernd Kiefer, Antoine Cully, Oya Celiktutan, Bert Bierman, Rifca Rijgersberg-Peters, Joost Broekens, Willeke van Vught, Michael van Bakkum, Yiannis Demiris, and Mark A. Neerincx. 2021. A cloud-based robot system for long-term interaction: Principles, implementation, lessons learned. *ACM Transactions on Human-Robot Interaction* 11, 1 (October 2021), Article 8. <https://doi.org/10.1145/3481585>